Why do durations in musical rhythms conform to small integer ratios?

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All authors listed have made substantial, direct, and intellectual contributions to the work, and approved it for publication.

Keywords

Rhythm, music perception, Neural oscillations, Integer ratio, Scalar Expectancy theory

Abstract

Word count: 250

One curious aspect of human timing is the organization of rhythmic patterns in small integer ratios. Behavioral and neural research showed that adjacent time intervals in rhythms tend to be perceived and reproduced as approximate fractions of small numbers (3/2 rather than 23/51). Recent work on iterated learning and reproduction further supports this: when given a randomly timed drum pattern to imitate, participants subconsciously transform it toward small integer ratios. The mechanisms accounting for this 'attractor' phenomenon are not fully understood, but two theoretical frameworks may offer an explanation. Oscillatory theories suggest it to be a by-product of integer ratio rhythms in oscillatory activity between neural populations. The scalar expectancy theory explains time perception and production using Weber's law: just detectable durational differences between time intervals are a constant fraction of their duration. In this piece, we put forward the hypothesis that the integer-ratio property found in rhythm experiments is compatible with a combination of the two frameworks. Specifically, we draw on the scalar property of timing to formulate a simple model of categorical perception that may result in an integer-ratio bias. We support our integrative approach with mathematical derivations to make assumptions more explicit and provide testable predictions. We present equations to calculate predicted durational ratios based on minimal assumptions on: (i) one ratio between two durations, (ii) a scalar timing constant, and (iii) the amount of overlap between durational categories. Our derivations provide the basis for future computational and behavioural work to expand, or refute, our hypothesis empirically.

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Ethics statements

(Authors are required to state the ethical considerations of their study in the manuscript, including for cases where the study was exempt from ethical approval procedures)

Does the study presented in the manuscript involve human or animal subjects: No

Why do durations in musical rhythms conform to small integer ratios?

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Perspective Article for the Research Topic 'Temporal Structure of Neural Processes Coupling Sensory, Motor and Cognitive Functions of the Brain'

Current word count: 3571-311-47=**3213**

Short title: **Why do rhythms have integer ratios?**

Abstract (max 250, currently 248)

One curious aspect of human timing is the organization of rhythmic patterns in small integer ratios. Behavioral and neural research showed that adjacent time intervals in rhythms tend to be perceived and reproduced as approximate fractions of small numbers (3/2 rather than 23/51). Recent work on iterated learning and reproduction further supports this: when given a randomly timed drum pattern to imitate, participants subconsciously transform it toward small integer ratios. The mechanisms accounting for this 'attractor' phenomenon are not fully understood, but two theoretical frameworks may offer an explanation. Oscillatory theories suggest it to be a by-product of integer ratio rhythms in oscillatory activity between neural populations. The scalar expectancy theory explains time perception and production using Weber's law: just detectable durational differences between time intervals are a constant fraction of their duration. In this piece, we put forward the hypothesis that the integer-ratio property found in rhythm experiments is compatible with a combination of the two frameworks. Specifically, we draw on the scalar property of timing to formulate a simple model of categorical perception that may result in an integer-ratio bias. We support our integrative approach with mathematical derivations to make assumptions more explicit and provide testable predictions. We present equations to calculate predicted durational ratios based on minimal assumptions on: (i) one ratio between two durations, (ii) a scalar timing constant, and (iii) the amount of overlap between durational categories. Our derivations provide the basis for future computational and behavioural work to expand, or refute, our hypothesis empirically.

Keywords: rhythm, music perception, scalar expectancy theory, neural oscillations, integer ratio

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Integer ratios and musical rhythm

What are small integer ratios? A *ratio* between two inter-onset intervals (IOIs) is the division between two, usually adjacent durations. *Integer* ratios can be written as a fraction: 1.5 equals 15/10 or 3/2, but $\sqrt{2}$ for instance cannot be written as a fraction. An integer ratio is *small* if the result of the division can be written as a small integer number divided by another small integer number e.g. 2/3, but not 23/51 (Pikovsky et al., 2003; Strogatz, 2003). Why are integer-ratio rhythms special?

Integer-ratio auditory rhythms are common in music (Essens and Povel, 1985; Savage et al. 2015; Toussaint, 2013). Behavioral and neural research point to a fine-grained representation of rhythms formed with small integer ratios. For instance, for those rhythms, individuals tend to show improved deviance detection (Jones & Yee, 1997; Large & Jones, 1999), enhanced memory (Deutsch, 1986; Palmer & Krumhansl, 1990) and reproduction (Povel & Essens, 1985; Essens, 1986), and better synchronization (Patel et al., 2005), when compared to non-metrical rhythms, i.e., rhythms organized in larger values (e.g., 1:5) or non-integer values (e.g., 1:2.5.). The distortion of near-integer ratios toward integer ones (or their harmonics) reported in behavioral (Fraisse, 1982) and neurophysiological studies (Motz et al., 2013) further supports a preference for integer ratio rhythms: this bias may be a 'cognitive attractor'. This idea has recently received support from studies of iterated learning and reproduction. When humans copy an initially random rhythmic sequence, and this imitation process is repeated in a cascade fashion, the sequence is rapidly and subconsciously reshaped to be composed of IOIs related by small integer ratios (Figure 1A; Jacoby & McDermott, 2017; Ravignani et al. 2016, 2018, c.f. Polak et al., 2016).

Why do these small integer ratio have a privileged role in musical rhythms? Why are humans drawn to rhythms with such a peculiar mathematical property? Does this property reflect a special quirk of music perception and motor sequencing, or could it be predicted from other, domain-general aspects of cognition? Can we express these alternatives mathematically? How could we test these alternatives experimentally? We explore mathematically the possibility that the human bias for intervals related by simple integer ratios can be understood as deriving from a combination of other previously unconnected features of perception.

We begin by outlining some classical frameworks for human timing. We then summarize the evidence that supports a human bias towards rhythmic intervals characterized by integer ratios. We then present our proposal for linking timing frameworks and integer ratios through mathematical formalisms. Specifically, we draw on the scalar property of time interval estimation to formulate a simple model of categorical perception that may result in an integer ratio bias (Figure 1). We conclude with a brief discussion of the principle merits and limitations of our hypothesis, and outline future goals.

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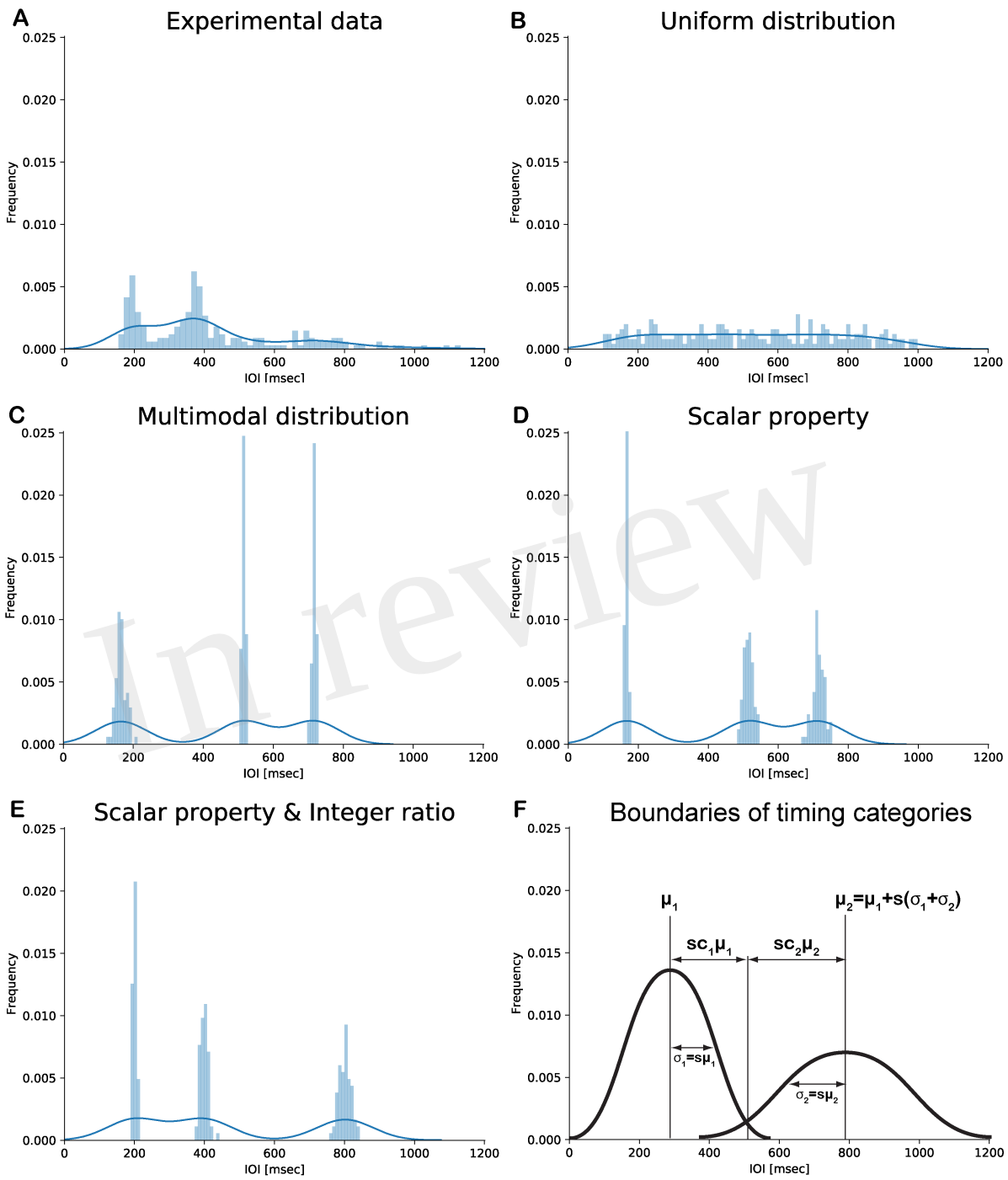

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Figure 1. Graphical representation of different types of IOI distributions. (A) Empirical distribution of drumming data showing integer ratio categories. Data from the last experimental generation of chain 2 in Ravignani et al. (2016). (B) Uniform distribution from 100 to 1000 ms. (C) Multimodal distribution based on 3 randomly chosen centroids without further assumptions. (D) Multimodal distribution around the same 3 centroids assuming the scalar timing property. (E) Multimodal distribution assuming the scalar timing property and showing small integer ratios. Data in panels (B-E) was randomly sampled from normal distributions, so that N equalled the number of samples in panel A. (F) Schematic representation of quantitative constraints linking scalar timing and small integer ratios. Panel F was produced without simulated or experimental data. We realise that categorical timing is a psychological concept (Desain & Honing, 2003; Drake & Bertrand, 2001; Schultze, 1989; ten Hoopen et al, 2006), based on behavioral observations and a simplification of the more complex underlying neural reality. However, notice how the notion of a

durational category may not be that far off neural reality. For instance, categories can be mapped to interval tuning in the premotor neurons of monkeys performing a synchronization continuation task (Merchant et al., 2013). Here, the distribution of preferred intervals could be viewed as a prior, although this distribution is uniform, rather than bimodal as in (Merchant et al., 2013). Human neuroimaging work has shown specific activation patterns for the perceptual processing of integer interval ratios (Sakai et al., 1999). Note that, small integer ratios may or may not compose a metrical beat structure give rise to a metrical percept. A metrical beat can be created by regularly spaced accents over time, that occur at constant multiple integer of the smallest time unit/category (Povel & Essens, 1985). In other words, metrical structure presents a second-level demonstration of the psychological prior of small-integer ratios.

Psychophysical and oscillatory frameworks

Several theoretical frameworks have been suggested to account for the mechanisms behind human timing (Allman & Meck, 2011; Allman et al., 2014; Church, 1999; Getty, 1975; Grondin, 2001,2010; Ivry & Schlerf, 2008; Karmarkar, & Buonomano, 2007; Mauk and Buonomano, 2004; Meck, 1996; Merker, 2014; Wearden, 1991; Wing & Kristofferson, 1973a, 1973b). Historically, the 'scalar expectancy theory' has been an influential theoretical framework in psychophysics (Allman & Meck, 2001; Gibbon, 1977; Wearden, 1991). The scalar property of timing follows Weber's law (Bizo et al., 2006): the error of an interval duration being timed is proportional to the mean duration of that interval. One, perception-based formulation states that the ratio between the just noticeable difference and the duration of a reference stimulus is constant across stimulus length (Grondin, 2001). In the other formulation, the coefficient of variation (standard deviation divided by mean) in estimating durations is constant across durations (Figure 1D; Gibbon, 1977). Scalar expectancy theory is often presented together with the concept of an accumulator, explicitly tracking temporal units of events and comparing them (Matell and Meck, 2000; Allman et al., 2014). A different framework for timing mechanisms coming from neuroscience and physics suggests that neural oscillations entrain (or 'resonate') with the periodicity of external stimuli at multiple time-scales (Arnal & Giraud, 2012; Aubanel et al., 2016; Buzsaki, 2006; Celma-Miralles et al., 2016; Large, 2008). Specifically, it predicts that the phase and frequency of neural oscillations are tuned to the phase and frequency of external events at multiple metrical levels. For instance, processing of a metronome beat will induce low-frequency oscillations and/or power fluctuations in high-frequency oscillations following the beat frequency, their multiples, or their divisors. Critically, the stability of the connection between two or more active neural oscillations, i.e., the 'resistance' to external perturbations, depends on the ratio of their periods (e.g., 1:1, 2:1, 2:3). Small integer ratios typically confer greater stability. This may explain the perceptual advantage for integer-ratio stimuli over more complex metrical patterns (Large and Kolen, 1995). Other frameworks state that specific neurons or neural channels are tuned to particular durational intervals (Merchant et al., 2013; Bartolo et al., 2014).

Iterated drumming experiments: Small integer ratios as cognitive attractors

Recent behavioural research investigated human priors on durations in rhythmic patterns (Jacoby & McDermott, 2017; Ravignani et al. 2016,2018). Participants were given drumming sequences to imitate to the best of their abilities. The patterns produced were presented to the same or new participants to imitate in an iterative procedure. Strikingly, 'first-generation' participants were given completely random patterns, yet 'last-generation' participants produced rhythms exhibiting small integer ratios, a universal property of human musical cultures. These

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recent results dovetail with previous work on e.g. bimanual tapping (Peper et al., 1991a,1991b; Peper et al., 1995; Peper and Beek, 1998)

More specifically, participants were presented with sequences of IOIs sampled from a uniform distribution *U* (e.g. Figure 1B). As the patterns were transmitted through 'chains of participants' (Jacoby & McDermott, 2017; Ravignani et al. 2016,2018), *U* converged towards a distribution *D*, which can be understood to reflect a human observer's posterior distribution of IOIs (e.g. Figure 1A). This distribution is multimodal and the modes are related by small integer ratios (Jacoby & McDermott, 2017; Ravignani et al. 2016).

Here we address the following question: Is it possible to explain the small-integer ratios in *D* via established psychophysical principles, none of which explicitly entail them? Equivalently, is the integer ratio bias a perceptual primitive, or does it derive from more fundamental primitives? Jacoby & McDermott (2017) related a theoretical prior, where integer ratios were built in, to an empirically estimated prior, showing that these aligned. Here, we investigate whether it is possible to derive a prior with similar properties by not building in the integer-ratio, but by combining empirically-motivated general principles of timing with weak assumptions which can be refined by future testing.

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Probabilistic Inference for Interval Ratio Categories

Conceptually, our question is: Under which conditions will *G* show small-integer ratios, without having built these ratios into the model?

Without any assumption, *G* would equal the uniform IOI distribution *U in expectation*. Which psychophysical results on basic mechanisms of rhythm perception and production can allow us to turn *U* into *G*? Below, we make four assumptions, each based on psychophysical evidence. The number of free parameters in the model is drastically reduced with little loss of generality. We first elaborate on previous formalizations to make relevant assumptions explicit and comparable.

Assumption 1: Categorical timing

An *n*-interval rhythm defines a sequence of inter-onset-intervals $\boldsymbol{d} = (d_1, \dots, d_{n-1})$ and of ratios $\boldsymbol{r} = (r_1, \dots, r_{n-2})$, such that $r_i = d_{i+1}/d_i$. Perception of a rhythm $\boldsymbol{r}$ induces a representation $\boldsymbol{z} = (z_1, \dots, z_{n-2})$, with a strong tendency to categorise. The vector $\boldsymbol{z}$ is a sequence of a small number of mental/phenomenal interval-ratio *categories* that exemplify the observed data $\boldsymbol{r}$. More specifically, the notation $z_i = k$ identifies that interval ratio r_i belongs to phenomenal category k (Ravignani et al., 2018). Whilst we will not use $\boldsymbol{z}$ explicitly in our calculations, $\boldsymbol{z}$ formalises the categorical framework that is assumed throughout. This is the first key assumption: induction of rhythmic sequences recruits a categorical interpretation of time intervals (Desain & Honing, 2003). This assumption is shared by previous approaches (Jacoby & McDermott, 2017), and related to basic durational tuning properties of premotor neurons as shown in non-human primates (Merchant et al., 2013). Behavioural evidence shows that human timing is similarly categorical: participants tapping, even without a reference pulse, produce IOI distributions with several distinct peaks reflecting underlying *categories* (Collyer et al., 1994). This suggests that

the distribution *G* can be approximated as a multimodal mixture of normal distributions (Figure 1C), rather than a uniform distribution (Figure 1B). This assumption does not constrain the mean of each category.

Assumption 2: Bayesian inference

Rhythm experiments often implicitly rely on the assumption that perceptual timing can be described as a process combining subjective prior beliefs with sensory input. One way to capture this assumption mathematically is to model perception as Bayesian inference (Merchant et al., 2013; Jazayeri & Shadlen 2010; Cicchini et al., 2012; Perez & Merchant 2018). This assumption is not absolutely necessary for our analysis here: what counts is the nature of the prior, rather than how it is deployed during perceptual interpretation. The Bayesian assumption provides a natural and useful framework in which to express a prior distribution as an inductive bias (Thompson et al., 2016), and has been fruitfully adopted in previous models of time interval estimation (e.g. Cicchini et al, 2012, Jazayeri and Shadlen 2010). In this framework, we can characterize participant behaviour as adopting a categorical representation for interval ratio r_i according to the distribution $p(z_i = k|r_i) \propto p(r_i|z_i = k)p(z_i = k)$. Our focus is the prior distribution over categories, $p(z_i = k)$, which we call *G*. Alternatively, it is also possible to model learner's assumptions about a likelihood distribution as a source of bias (e.g. Jazayeri and Shadlen 2010, Cicchini et al., 2012; Merchant et al., 2018).

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Assumption 2: Formulating a Prior Over Interval Ratio Categories

Other approaches model individual *n*-interval rhythms as single points in the *n-1* dimensional simplex, and formulate a multivariate-mixture prior over this space, assuming Gaussian models underlie each of the the mixture's components (Jacoby & McDermott, 2017). In other words, they formulated a multivariate $p(z)$ directly. Our formulation of the prior is closely related. Like Jacoby and McDermott (2017), we express the prior as a mixture of Gaussian components. However, our formulation treats an *n*-interval rhythm as a set of *n-1* independent samples from a *univariate distribution*, rather than a single multivariate sample. The two approaches essentially represent minor variants of the assumption concerning the covariance of interval ratio categories. The assumption that the distribution $p(z)$ has a Gaussian form is an approximation that should be tested in future work, but is in line with existing approaches to the topic and a fair first approximation which allows us to gain traction through known properties of the distribution.

We write the prior as a *K*-dimensional Gaussian mixture of interval ratio categories, and the data likelihood as i.i.d. Gaussian underlying these categories, such that the marginal distribution of interval ratios has the form:

$$p(\boldsymbol{r}) = G(\boldsymbol{r}) = \prod_{i=1}^{n-2} \sum_{k=1}^{K} \varphi_k N(d_i; \mu_k, \sigma_k) \quad (1)$$

Here, the prior assigns to each Gaussian *k=1,...,K* a weight in the mixture, φ_k, which determines its relative prominence as a category; a category mean μ_k, which specifies the expected interval ratio underlying this category; and a category variance σ_k. Our default assumption is that

weights are uniform: $\varphi_k = K^{-1}$. Whilst we hope to examine this assumption empirically in the future, we proceed under the most neutral assumption: no interval-ratio category is privileged.

Assumption 3: A small number of sub-second categories

Assume that our indexing of categories under the prior is strictly ordered by the category centres, such that $\mu_j < \mu_k \Leftrightarrow j < k$. We can immediately express our second empirical constraint on *G*: there are few categories (Desain & Honing, 2003; Motz et al., 2013; Ravignani et al., 2016, 2018). In other words, *K* is limited by our aim to only model components for *small* integer ratios, and these are limited in number. Furthermore, we know that category means μ_k are approximately bounded from below by 200 ms (London, 2004) above by 1000 milliseconds (Buhusi & Meck, 2005). This constraint bounds *K* to the largest number of categories such that no category mean exceeds 1 second, namely:

$$K = argmax_k \; s.t. \, \mu_k \leq 1000 \text{ for } k = 1, \dots, K \; . \tag{2}$$

So far, our assumptions constrain neither category means μ_k nor standard deviations σ_k.

Assumption 4: Scalar Timing

Our last assumption, which is perhaps the central contribution, is to note that timing is thought to exhibit *scalar properties*, at least within the range of sub-second time intervals used in the experiments considered here (Friberg & Sundberg, 1995; Gibbon, 1977; Madison & Merker, 2004; Matell and Meck, 2000). Scalar timing drastically reduces the number of free parameters describing *G*, by expressing category variances as a function of category means. The standard deviation of each category σ_k equals the mean μ_k multiplied by a constant factor *s* (Figure 1E):

$$\sigma_k = s \, \mu_k. \tag{3}$$

Previous empirical literature has shown *s* approximates *0.025* (Friberg & Sundberg, 1995; Madison & Merker, 2004).

Linking Integer Ratio Intervals to Scalar Timing: How close can we get?

The constraints considered so far are all empirically and independently motivated. *G* can be further characterized by the degree of overlap between Gaussians composing a Gaussian mixture. To formalize this notion, we assume each category *k* to intersect with its adjacent neighbours *k-1* and *k+1* at a distance c_k away from its mean μ_k, which is a constant multiple of the standard deviation σ_k. c_k parameterizes the overlap between categories: it expresses how many standard deviations away from its mean μ_k the cluster *k* intersects the cluster *k+1* (Figure 1F shows an example for k=1,2).

Combining this idea of a parameterized overlap with the scalar property, each cluster *k* extends from $\mu_k - sc_k\mu_k$ to $\mu_k + sc_k\mu_k$. Under these assumptions, the distance between the means of two adjacent distributions (Figure 1F) can be written as

$$\mu_{k+1} - \mu_k = sc_{k+1}\mu_{k+1} + sc_k\mu_k \; , \tag{4}$$

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while their ratio as

$$r_k = \mu_{k+1}/\mu_k \ . \tag{5}$$

Substituting (5) into (4), we obtain

$$r_k\mu_k - \mu_k = sc_{k+1}r_k\mu_k + sc_k\mu_k \ , \tag{6}$$

which can be simplified and rewritten as

$$r_k = (1 + sc_k)/(1 - sc_{k+1}) \ . \tag{7}$$

Equation (7) requires, to be well defined, that

$$0 < c_k < \frac{1}{s} \ , \tag{8}$$

which translates into a constraint on centroids to neither overlap nor be further than ≈ 40 standard deviations away from each other.

Operationally, the category centres following from the constraints on *G* can be calculated using the recursion equation:

$$\mu_{k+1} = r_k\mu_k \ . \tag{9}$$

The constraints structure the space of component Gaussians in the prior such that, by specifying μ_1, we can compute μ_k for all *k*≤*K* using equation (9)(Figure 1E).

These quantitative tools enable formulation of several questions. Given our post-hoc knowledge that the prior is characterized by categories centred at small multiple integer ratios, do the constraints we laid out structure the prior such that systematic integer-ratio clusters are predicted by setting μ_1 to the smallest possible integer ratio? An alternative approach might be to assume that one ratio is e.g. ½, and ask whether our equations imply integer ratios for the remaining cluster centres. More generally, do the constraints laid out impose an integer ratio structure on the prior without assuming an integer ratio for any of the clusters, simply by setting c_k in a certain way?

The special case of a constant c_k

Note that for $c_k = c$, most equations above lose dependence on *k*. If c_k is constant across categories, this is equivalent to assuming that each category contains the same proportion of the probability density function.

For c_k constant, equation (9) becomes

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$$\mu_{k+1} = r\mu_k \quad , \tag{10}$$

meaning constancy of adjacent ratios, or

$$\mu_k = r^{k-1}\,\mu_1 \quad , \tag{11}$$

the generating function providing the location of the kth centroid as a function of k and μ_1 alone.

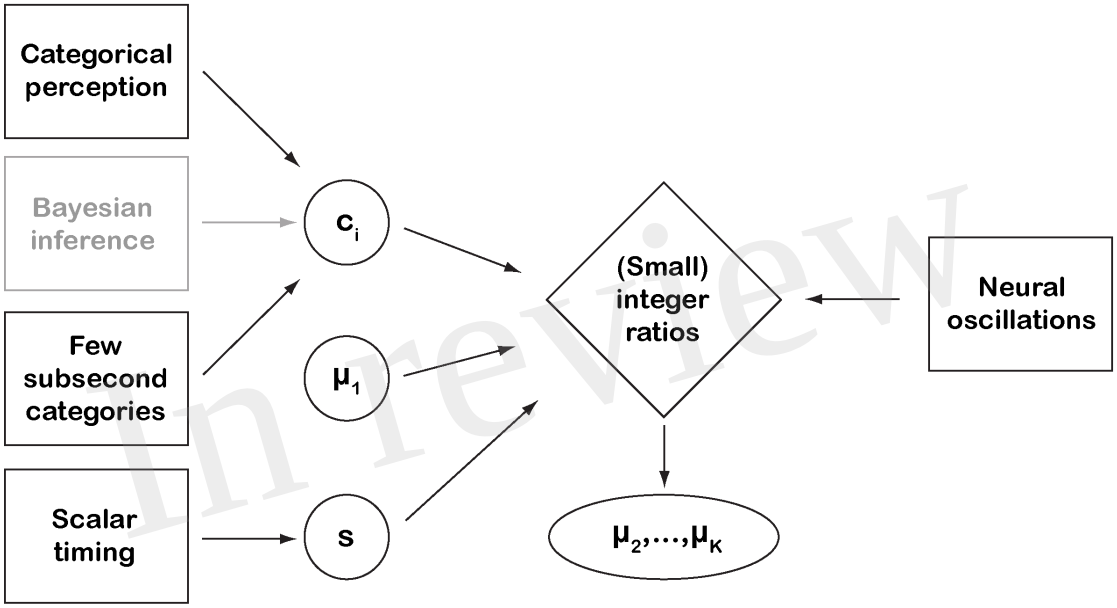

Figure 2. Schematic representation of the perspective introduced by this paper. 'Bayesian inference' is shaded in grey to mean that it is used here as a general working assumption and conceptual framework, rather than a strong, empirically supported assumption on cognitive processes (Shi, Church & Meck, 2013).

Suggested experiments: Modelling and Psychophysics

Equations (7) and (9) provide a potential link between scalar timing and integer ratios (Figure 2). These generative formulas can be implemented in computational simulations to explore the shape of the parameter space. Given specific values for parameters *s* and c_k, the equations will provide a unique set of ratios: are these small integer ratios? Likewise, given one single integer ratio μ_1, all other μ_k are determined by equation (9): which values of μ_1 result in *r* being integer ratios and psychophysically plausible values of *s* and c_k?

The perspective we offer here creates the basis for expanding into theoretical and empirical work on c_k and *s*. In particular, experimental research could advance this approach by estimating c_k and *s* via equation (7). In our model, the parameter *s* has been treated as a one-valued variable, whose mean is known (*s=0.025*). To make the model more realistic, it would be important to infer the distribution, hence the variance, of *s*. The distribution of *s* can be found by repeating previous psychophysical experiments such as those by Friberg & Sundberg (1995)

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and Madison & Merker (2004). Actual values of c_k can be estimated from experimental data by testing the perception and misattribution of durational categories. This way one could also test whether c_k is constant, or a function of other variables.

Limitations, discussion, and conclusions

The proposed way of representing rhythmic structure depends, among other factors, on the constancy of r_k. A deviation from this constancy would result in larger integer ratios, with the deviation accumulating over the categories when iterating equation (8). Empirical work (e.g. Ravignani et al., 2016; Jacoby & McDermott, 2017) has approached the connection between the 'mathematical perfection' of integer ratios and their empirical counterpart in a number of alternative ways. This perspective does not address how and when a real number is perceived as an integer ratio, leaving this as an empirical question for psychophysics research. In general, large integer ratios, and even irrational-number ratios, can be perceived as small integer ratios if close enough to one. For instance, $2^{7/12}\approx 1.498307$ is irrational (Coxeter, 1968) but close to 3/2. Virtually all pianos, today, employ this irrational number (1.498307) in their well-tempered tuning, which is 'close enough' for human hearing to the integer ratio 3:2.

In this short paper, we explore quantitative links between scalar timing, human perceptual biases, and integer ratio rhythms. The arguments we provide restrict the explanatory space to few competing alternative hypotheses. One possibility is that integer ratios should not be assumed as a human cognitive primitive. Instead, they could simply be a neurobiological by-product of, and mathematically derived from, other cognitive constraints.

Alternatively, it might be that the whole scalar timing framework is not the most suited to explain the integer-ratio phenomenon of human timing. If one adopts oscillatory frameworks, integer ratios swiftly derive, while scalar properties and categorical perception may be by-products of the oscillatory nature of brain activity. In particular, small integer ratios might just be an epiphenomenon of harmonics of one oscillator or interaction of two oscillators (Buzsaki, 2006; Collyer et al., 1994; Merker, 2014; Strogatz, 2003). Neural resonance to musical rhythm (Large, 2008), interval tuning (Merchant et al., 2013; Bartolo et al., 2014), and population clocks (Bakhurin et al., 2016; Gouvea et al., 2015; Crowe et al., 2014; Merchant and Averbeck, 2017) are alternative timing mechanisms, and have the advantages of stemming from in-vivo recordings of neural populations and being compatible with small integer ratios.

In any case, scalar timing and oscillatory theories are simplifications, i.e. approximate descriptions bound to specific experimental set-ups. At a neural level, the difference between scalar timing and oscillatory theories is more complex than it may appear to higher level cognitive theories. Only detailed neural models will enable us to capture the real underlying timing mechanisms.

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Conflict of Interest Statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Author Contributions

All authors listed have made substantial, direct, and intellectual contributions to the work, and approved it for publication.

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